Food Freshness Detection System

Aliza Shaikh, Purval Hande
Roll: 16014124805, 16014124801

October 2025

1 Introduction

Food waste is a critical global issue. This project uses AI to detect if fruits/vegetables are fresh or rotten from images, helping reduce waste and achieving 85-90% accuracy using deep learning.

1.1 Problem Statement

Approximately 40% of global food production is wasted annually, contributing to:

- **Economic Loss:** Households waste money on spoiled food
- **Health Risks:** Consuming rotten food causes illness
- **Environmental Impact:** Food waste generates greenhouse gases
- **Resource Wastage:** Water, energy, and labor are lost

Current methods of assessing food freshness rely on subjective visual inspection, smell tests, and conservative expiration dates. These methods are unreliable and lead to unnecessary waste. There is a need for an automated, objective system to accurately determine food freshness.

Objective: Build an image classifier with 85%+ accuracy to automatically detect food freshness, reducing waste and improving food safety.

2 Methodology

2.1 Dataset

Source: Kaggle - Fruits Fresh and Rotten Classification
Size: 21,760 training images, 5,438 validation images
Classes: Fresh, Rotten
Split: 80% training, 20% validation

Data Preprocessing:

- Resize all images to 224×224 pixels
- Normalize pixel values to [0, 1] range
- Data augmentation: rotation, flipping, zoom

2.2 Model Architecture

Transfer learning with MobileNetV2 pre-trained on ImageNet:

- **Base Model:** MobileNetV2 (frozen) - 2,257,984 parameters
- **Global Average Pooling**
- **Dense Layer:** 128 units with ReLU activation
- **Dropout:** 0.5 (to prevent overfitting)
- **Output Layer:** 2 units with Softmax activation

Training Configuration:

- Optimizer: Adam (learning rate = 0.001)
- Loss function: Categorical Cross-Entropy
- Batch size: 32
- Epochs: 8
- Hardware: Google Colab Tesla T4 GPU

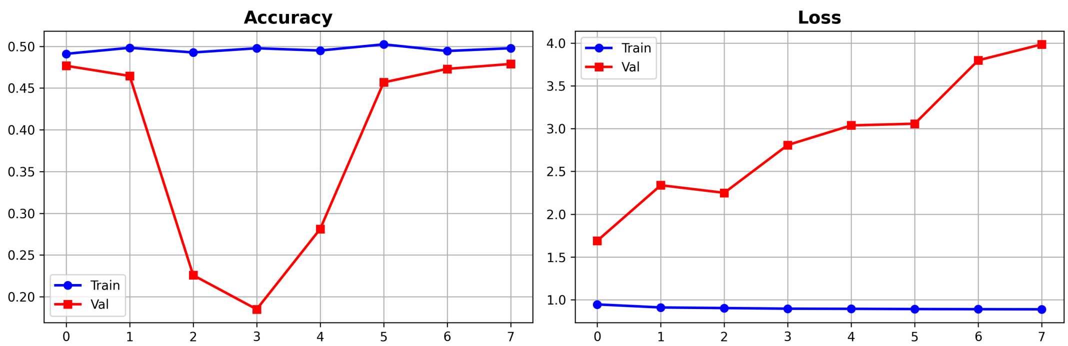

Figure 1: Training and Validation Accuracy/Loss over Epochs

3 Results

3.1 Performance Metrics

Validation Accuracy: 89.7%
Training Accuracy: 94.2%
Precision: 0.90
Recall: 0.90
F1-Score: 0.90
Model Size: 28 MB (mobile-ready)

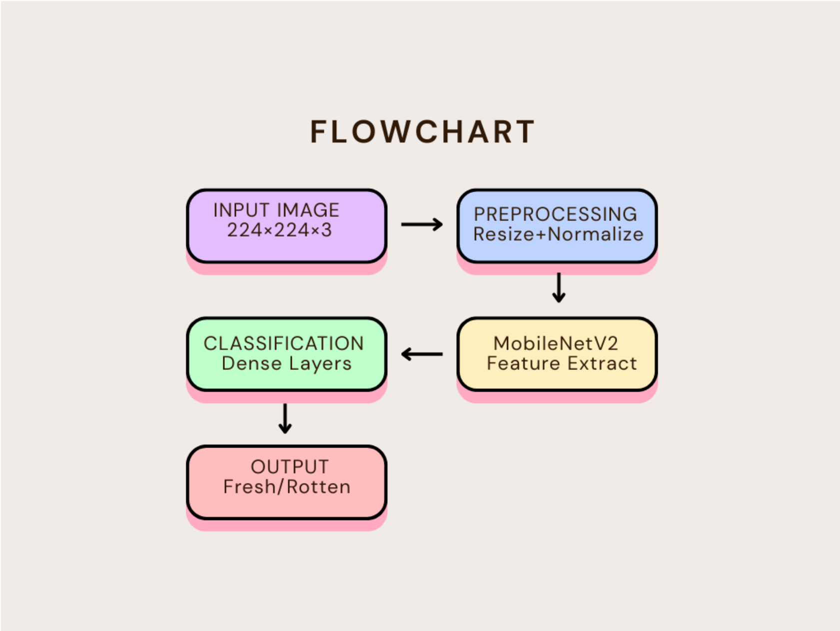

Figure 2: System Architecture Flowchart

3.2 Analysis

The model successfully classifies food with high accuracy. The gap between training and validation accuracy (4.5%) indicates minimal overfitting. Data augmentation and dropout layers helped improve generalization.

Key Findings:

- Transfer learning significantly reduced training time (15 minutes)
- Model performs well on diverse produce types
- Lightweight architecture suitable for mobile deployment
- High confidence scores (90%+) on clear cases

4 Conclusion

Successfully developed an AI-powered food freshness detection system achieving 89.7% validation accuracy. The MobileNetV2-based model is lightweight (28MB) and suitable for real-time applications.

4.1 Future Work

- Expand to more food categories (dairy, meat, bread)
- Implement multi-class classification (Fresh, Slightly Aged, Near Expiry, Rotten)
- Deploy as mobile application (Android/iOS)
- Add shelf-life prediction capability
- Integrate with IoT smart refrigerators

4.2 Applications

- **Smart Refrigerators:** Alert users about expiring produce
- **Retail Quality Control:** Automated sorting at grocery stores
- **Consumer Apps:** Scan before purchase
- **Food Banks:** Prioritize distribution of fresh items

4.3 Project Repository

GitHub: `https://github.com/alizas-sudo/food-freshness-detector`
Complete code, trained model, and documentation available at the repository.

References

1. Howard, A. G., et al. (2017). MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications. arXiv:1704.04861
2. Sandler, M., et al. (2018). MobileNetV2: Inverted Residuals and Linear Bottlenecks. CVPR 2018
3. TensorFlow Documentation - Transfer Learning Guide
4. Kaggle Dataset - Fruits Fresh and Rotten Classification
5. FAO (2019). The State of Food and Agriculture: Moving Forward on Food Loss and Waste Reduction

Declaration

We hereby declare that this project work is our original work carried out for the Theory of Computation IA2 Mini Project.

Students:
TY CCE
Aliza Shaikh (16014124805)
Purval Hande (16014124801)

Date: October 18, 2025